

Loops Part 2

Day 64
12/09/2025

Why Understanding Loops is Important in Programming?

Loops are one of the foundational blocks of programming. They help programmers.

- Repeat tasks without rewriting the code
- Traverse arrays, strings, linked list, trees, graphs
- Simulate real world problems (banking systems, simulations, OS scheduling etc)
- Optimize solutions by reducing repetitive operations.

NOTE → If you don't fully understand loops, you will struggle with DSA because most algorithms boil down to some kind of looping pattern.

Types of loops

Most languages (Java, C++, Python) have :

1. **For loop** → Best you know how many times to run
2. **While loop** → Best when the end condition is unknown (like user input, or until a queue is empty)
3. **Do while loop** → Executes once before checking condition.

Patterns in loops (Some Examples)

When you start practicing coding, especially in beginner DSA, you'll find loop based patterns like

① Simple Counting

```
for (int i = 0 ; i <= 5 ; i++)  
    System.out.print (i) ;  
}
```

output : 0 1 2 3 4 5

② Star patterns (Pyramids, Triangles, Squares)

```

*
* *
* * *
* * * *
```

③ Number patterns

```

1
1 2
1 2 3
1 2 3 4
```

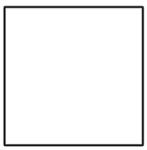
④ Character patterns

```

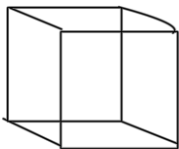
A
A B
A B C
A B C D
```

Nested Loops

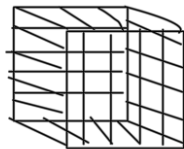
Nested loop is a loop inside another loop



- 2-level nested loops → print grids, 2D arrays, matrices



- 3+ level nested loops → used in some Brute force problems, though rare in optimized DSA



Using loops in DSA

Loops are the heart of Algorithms :

- Arrays → traversing, reversing, prefix sums.
- String → Palindrome checking, substring search
- Linked list → traversing nodes
- stack / queues → push/pop until empty/full

- Binary Trees $\longrightarrow$ BFS (queue + loop), DFS (stack / recursion)
- Graphs $\longrightarrow$ adjacency list traversal with nested loops,

DSA patterns & loops

Some classic loop based DSA patterns

1. Two pointers Techniques (loop with two moving indices)
2. Sliding Window (loop + shrinking / expanding window)
3. Nested Loops for Subarrays / Substrings
4. Matrix Traversal (row-wise, column-wise, diagonal)
5. Binary Search (loop with mid-point adjustments)

More About loops With Example

$\longrightarrow$ printing 100 stars in single line

* * * * * - - - - -

```

Loops > J Patterns.java > Java Language Support > Patterns >
1  package Loops;
2
3  public class Patterns {
4
5      public static void printStart1Line(int count){
6
7          for(int i = 0 ; i < count ; i++){
8
9              System.out.print(s:"* ");
10
11          }
12
13
14      public static void main(String[] args) {
15
16          printStart1Line(count:100);
17      }
18  }
19

```

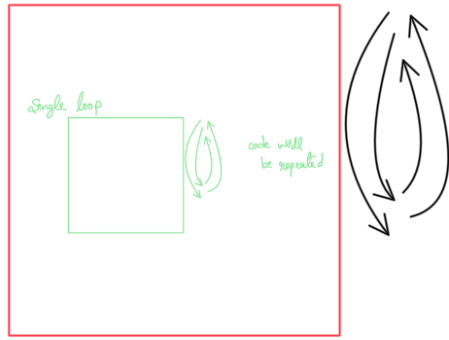
$\longrightarrow$ printing Hundred stars (10 stars at each line)

```

* * * * *
* * * * *
* * * * *
* * * * *
. . . . .
* * * * *

```


Single loop



code will be repeated

Nested loop runs repeatedly outer loop times

Nested loop → a loop inside a loop

```
public static void printGrid(int count){
    for(int i = 0 ; i < count ; i++){
        for(int j=0 ; j < count ;j++) {
            System.out.print(s:"* ");
        }
        System.out.println();
    }
    System.out.println();
}
```

Grid

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

To understand star pattern always think in the form of grid

*			
*	*		
*	*	*	
*	*	*	*

Rows
1
2
3
4

columns 1 2 3 4

*	*	*	*
*	*	*	*
*	*	*	*
*	*	*	*

Rows
1
2
3
4

columns 1 2 3 4

```
65
66 public static void gridPrintUsingArray(int [] nums ){
67
68     for(int i = 0; i < nums.length; i++){
69
70         for(int j = 0 ; j < nums[i] ; j++){
71
72             System.out.print( s:"* " );
73         }
74
75         System.out.println();
76     }
77 }
78
79
80 Run | Debug | Run main | Debug main
81 public static void main(String[] args) {
82     int nums[] = {2,4,5,7,6,8,10,3};
83
84     gridPrintUsingArray(nums);
85 }
```

```
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(base) kveeresh@Ks-Mac
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--enable-preview -XX:
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ation\ Support/Code/Us
f2b3509965b320b049bd62
s/Computer-Science_4e2
```

Tomorrow I will Explore More about loops and patterns printing.

See you tomorrow